FutureTech HackFest 2026

TrustRide

Governed & Safe Remote EV Immobilization

Every heist needs an inside job. We're stopping the one nobody's watching.

Kanneboina Shiva Kumar · Software Track — Cybersecurity Solutions

THE PROBLEM

A viral exploit exposed something bigger

In July 2026, an app called BAT-BMS went viral in India for one reason: **anyone within Bluetooth range could remotely cut power to a moving e-rickshaw.**

Cheap battery management systems accept unauthenticated BLE connections. The government banned the apps within days.

But the apps were originally legitimate tools — for financiers and fleet operators to disable a vehicle over unpaid loans or theft. The ban fixes the exploit. It doesn't fix the design.

3

apps pulled from app stores nationwide

0

authentication required by the exploit

10–15m

BLE range needed to disable a vehicle

THE REFRAME

Banning the app doesn't close the gap

WHAT GOT BANNED

An unauthenticated Bluetooth exploit anyone within 10–15 metres could trigger — no pairing, no PIN, no signature required. Surface-level: already patched by the nationwide ban, and would eventually be closed in firmware anyway as BMS vendors respond.

WHAT'S STILL BROKEN

Even a legitimate financier disabling a vehicle over unpaid loans has no motion-safety check, no cryptographic authorization, and no accountability trail proving who ordered it or why. That gap doesn't disappear when the news cycle does — it just goes back to being invisible.

Nobody — driver, financier, or regulator — currently has a safe, accountable way to remotely disable a vehicle.

OUR PROCESS

We didn't jump straight to building

1

Question the idea

Is this a real, lasting problem — or a viral moment that fades with the news cycle?

→

2

Find the real insight

The gap isn't better Bluetooth security. It's accountability for legitimate shutdown.

→

3

Test cheap before building

Battery teardowns and financier/driver interviews first — with clear stop-the-project triggers.

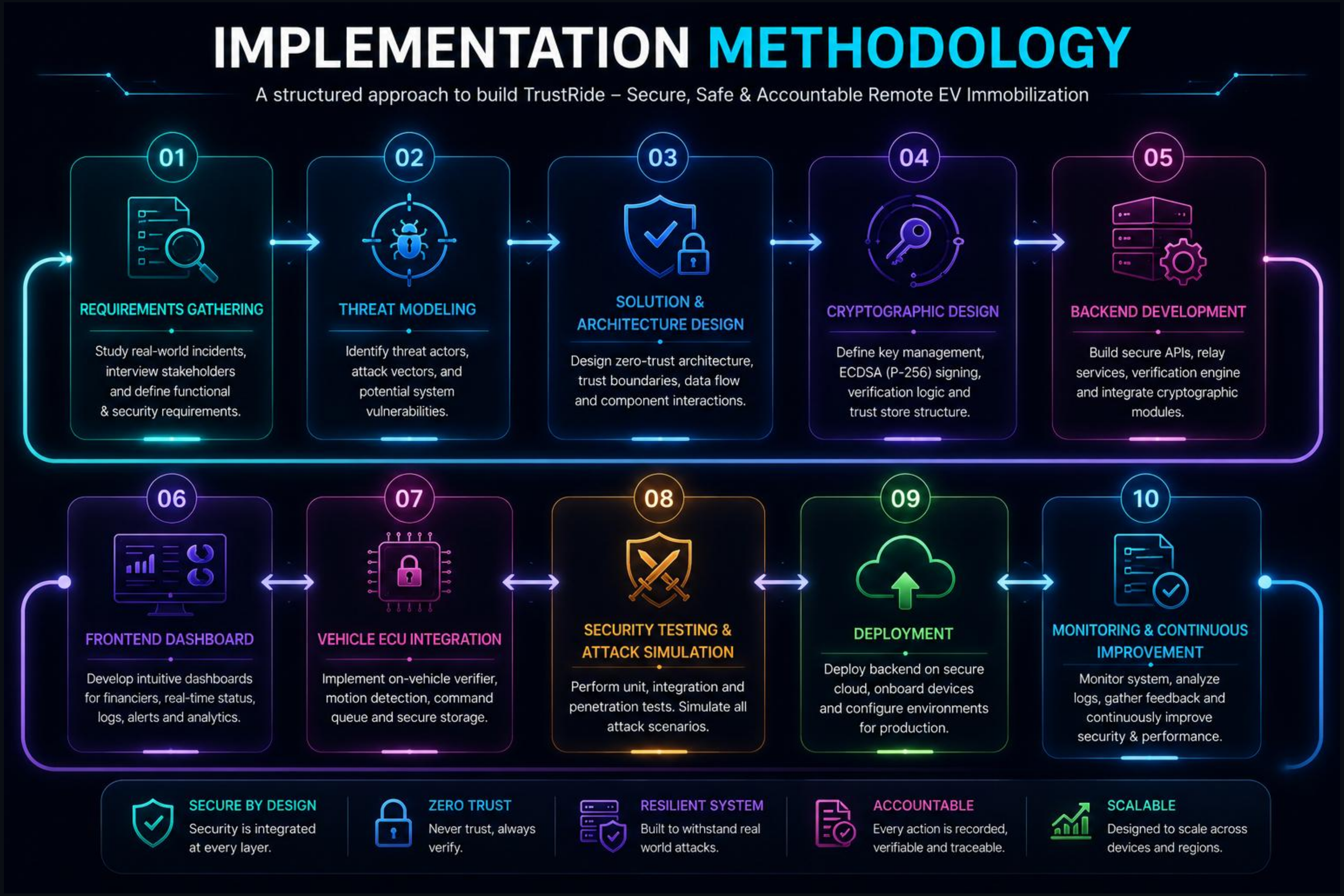
→

4

Build only after validation

This MVP: a working simulation, built to be shown to real financiers and drivers.

Only after this process did we build — not the other way around.


IMPLEMENTATION METHODOLOGY
A structured approach to build TrustRide – Secure, Safe & Accountable Remote EV Immobilization
01
REQUIREMENTS GATHERING
Study real-world incidents, interview stakeholders and define functional & security requirements.
02
THREAT MODELING
Identify threat actors, attack vectors, and potential system vulnerabilities.
03
SOLUTION & ARCHITECTURE DESIGN
Design zero-trust architecture, trust boundaries, data flow and component interactions.
04
CRYPTOGRAPHIC DESIGN
Define key management, ECDSA (P-256) signing, verification logic and trust store structure.
05
BACKEND DEVELOPMENT
Build secure APIs, relay services, verification engine and integrate cryptographic modules.
06
FRONTEND DASHBOARD
Develop intuitive dashboards for financiers, real-time status, logs, alerts and analytics.
07
VEHICLE ECU INTEGRATION
Implement on-vehicle verifier, motion detection, command queue and secure storage.
08
SECURITY TESTING & ATTACK SIMULATION
Perform unit, integration and penetration tests. Simulate all attack scenarios.
09
DEPLOYMENT
Deploy backend on secure cloud, onboard devices and configure environments for production.
10
MONITORING & CONTINUOUS IMPROVEMENT
Monitor system, analyze logs, gather feedback and continuously improve security & performance.
SECURE BY DESIGN
Security is integrated at every layer.
ZERO TRUST
Never trust, always verify.
RESILIENT SYSTEM
Built to withstand real world attacks.
ACCOUNTABLE
Every action is recorded, verifiable and traceable.
SCALABLE
Designed to scale across devices and regions.

OUR SOLUTION

Four pillars, not one patch

Motion-Safety Interlock

A shutdown command can never execute while the vehicle is moving — verified independently on the vehicle side, not self-reported.

Cryptographic Authorization

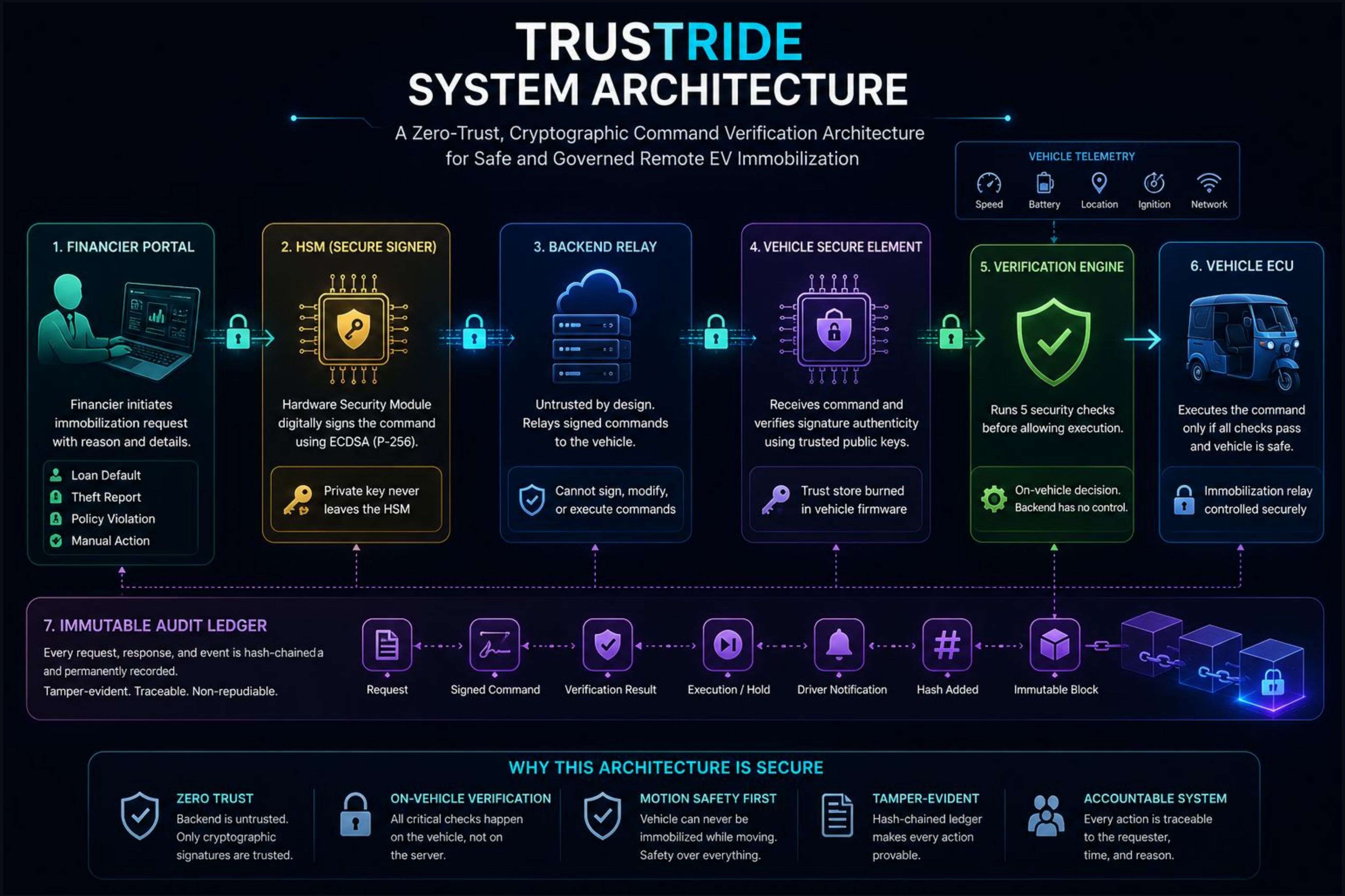
Every command is signed and verified with real ECDSA signatures. The backend cannot unilaterally execute a command — only a valid signature can.

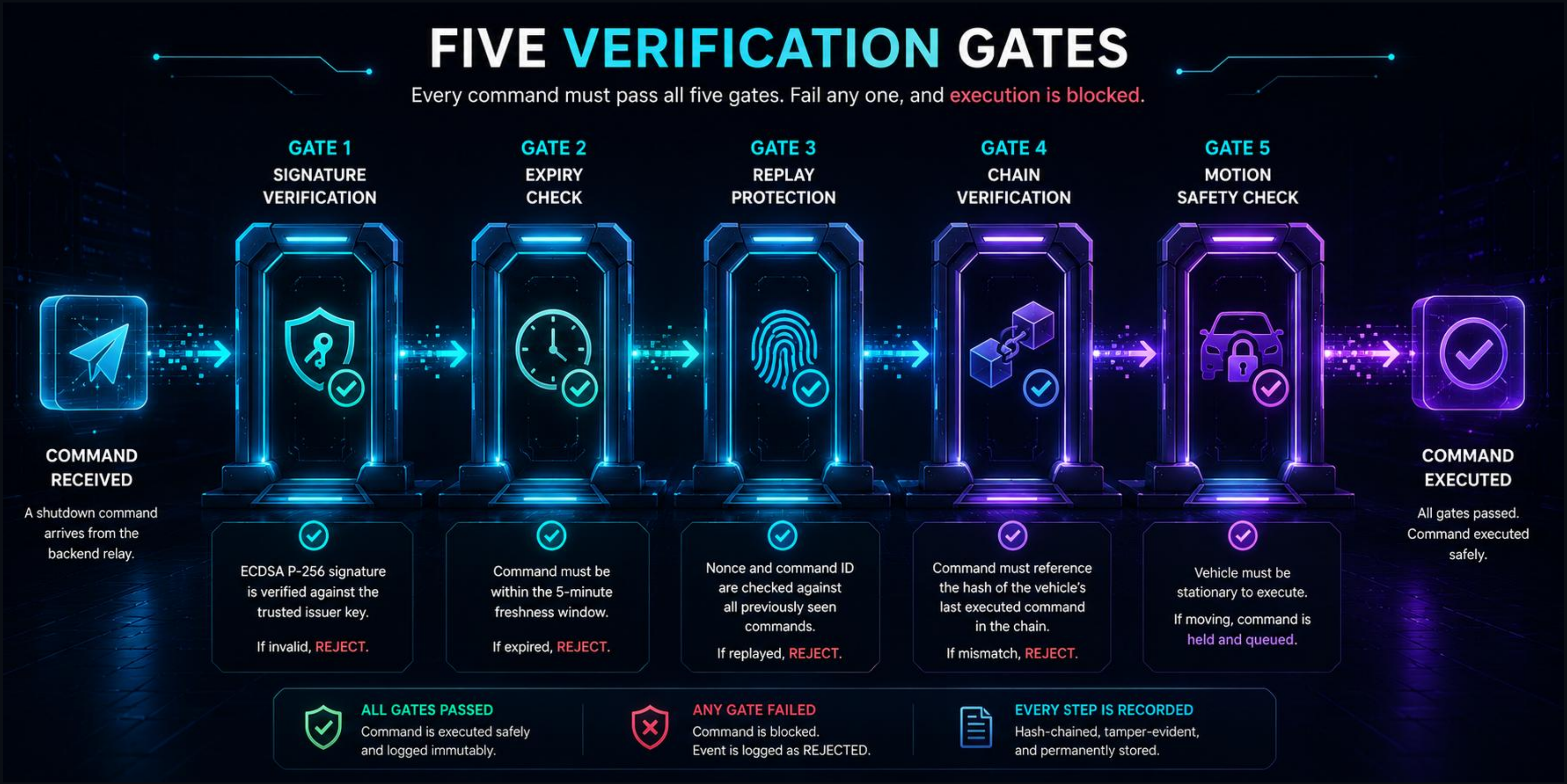
Tamper-Evident Audit Trail

Every request is hash-chained and permanently logged — who, when, why — provable later, not just claimed.

Multi-Signature Governance

High-risk commands now require dual signatures — Financier + Ops Admin — before the ECU accepts them.


TRUSTRIDE
SYSTEM ARCHITECTURE
A Zero-Trust, Cryptographic Command Verification Architecture
for Safe and Governed Remote EV Immobilization
VEHICLE TELEMETRY
Speed
Battery
Location
Ignition
Network
1. FINANCIER PORTAL
Financier initiates immobilization request with reason and details.
Loan Default
Theft Report
Policy Violation
Manual Action
2. HSM (SECURE SIGNER)
Hardware Security Module digitally signs the command using ECDSA (P-256).
Private key never leaves the HSM
3. BACKEND RELAY
Untrusted by design. Relays signed commands to the vehicle.
Cannot sign, modify, or execute commands
4. VEHICLE SECURE ELEMENT
Receives command and verifies signature authenticity using trusted public keys.
Trust store burned in vehicle firmware
5. VERIFICATION ENGINE
Runs 5 security checks before allowing execution.
On-vehicle decision. Backend has no control.
6. VEHICLE ECU
Executes the command only if all checks pass and vehicle is safe.
Immobilization relay controlled securely
7. IMMUTABLE AUDIT LEDGER
Every request, response, and event is hash-chaineda and permanently recorded.
Tamper-evident. Traceable. Non-repudiable.
Request
Signed Command
Verification Result
Execution / Hold
Driver Notification
Hash Added
Immutable Block
WHY THIS ARCHITECTURE IS SECURE
ZERO TRUST
Backend is untrusted. Only cryptographic signatures are trusted.
ON-VEHICLE VERIFICATION
All critical checks happen on the vehicle, not on the server.
MOTION SAFETY FIRST
Vehicle can never be immobilized while moving. Safety over everything.
TAMPER-EVIDENT
Hash-chained ledger makes every action provable.
ACCOUNTABLE SYSTEM
Every action is traceable to the requester, time, and reason.



FIVE VERIFICATION GATES
Every command must pass all five gates. Fail any one, and execution is blocked.
COMMAND RECEIVED
A shutdown command arrives from the backend relay.
GATE 1
SIGNATURE VERIFICATION
ECDSA P-256 signature is verified against the trusted issuer key.
If invalid, REJECT.
GATE 2
EXPIRY CHECK
Command must be within the 5-minute freshness window.
If expired, REJECT.
GATE 3
REPLAY PROTECTION
Nonce and command ID are checked against all previously seen commands.
If replayed, REJECT.
GATE 4
CHAIN VERIFICATION
Command must reference the hash of the vehicle's last executed command in the chain.
If mismatch, REJECT.
GATE 5
MOTION SAFETY CHECK
Vehicle must be stationary to execute.
If moving, command is held and queued.
COMMAND EXECUTED
All gates passed. Command executed safely.
ALL GATES PASSED
Command is executed safely and logged immutably.
ANY GATE FAILED
Command is blocked. Event is logged as REJECTED.
EVERY STEP IS RECORDED
Hash-chained, tamper-evident, and permanently stored.

AT A GLANCE

The adversarial gap, in one table

Solution	Motion Safety	Crypto Auth	Audit Trail	Adversarial Model
BAT-BMS / similar apps	✗ No	✗ No	✗ No	✗ No
Global fleet telematics	✓ Yes	⚠ Partial	⚠ Server logs	✗ No
Indian BMS-security startups	⚠ Physical only	✗ No	✗ No	✗ No
TrustRide	✓ Yes	✓ ECDSA (Dual-Sig)	✓ Hash-chain	✓ Yes

TrustRide is the only row with a full column of checkmarks — including the one nobody else even attempts.

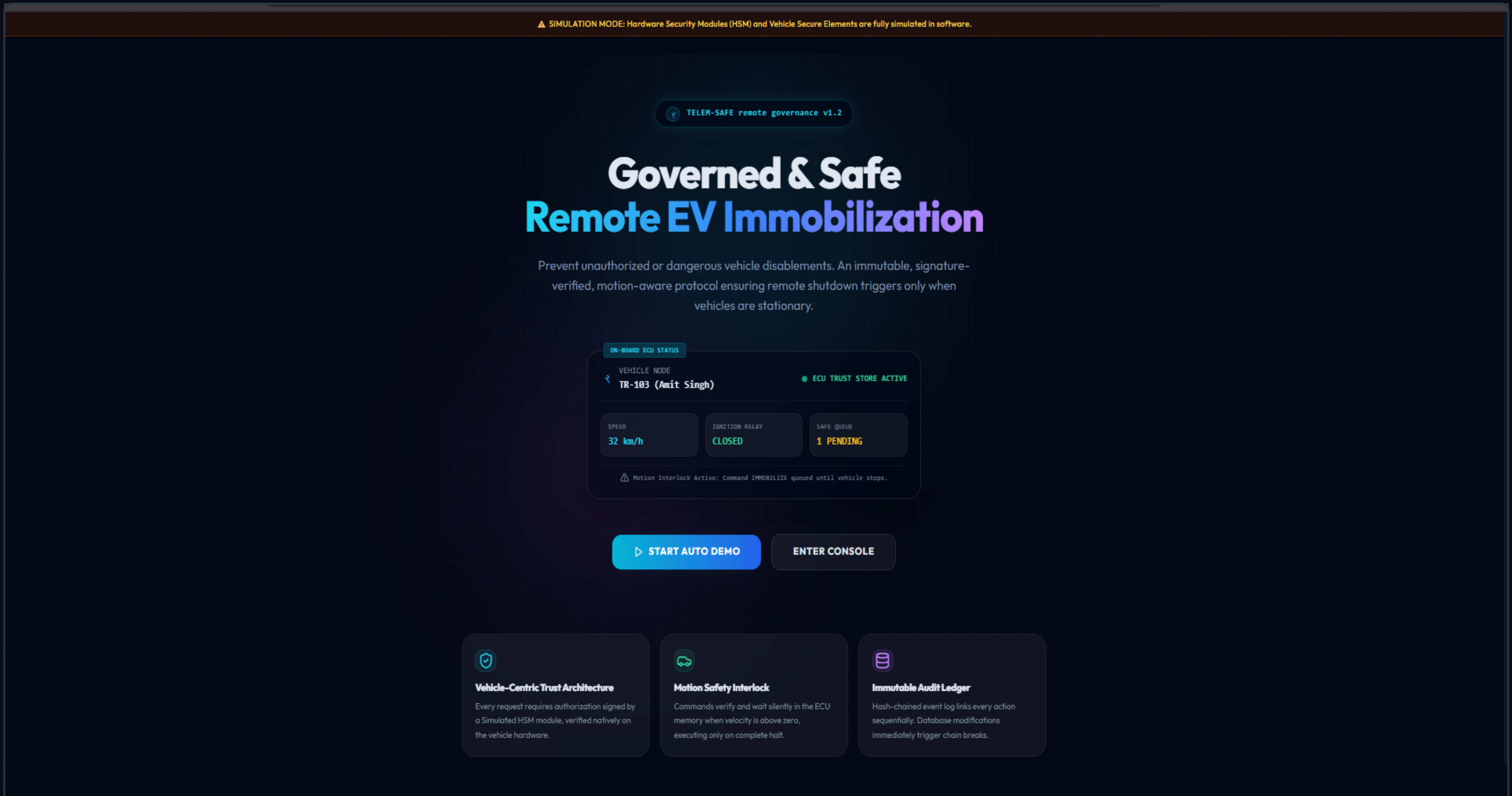
THREAT SANDBOX

Every attack vector, live-rejected

BLE Local Sweep	Check 1 — Signature	REJECTED
Man-in-the-Middle	Check 1 — Signature	REJECTED
Verbatim Replay	Check 3 — Replay	REJECTED
Stale Command Playback	Check 2 — Expiry	REJECTED
Rogue Issuer	Check 1 — Trust Store	REJECTED
Partial Signature Attack (1 of 2 Keys)	Gate 1 — Multi-Sig Threshold	REJECTED

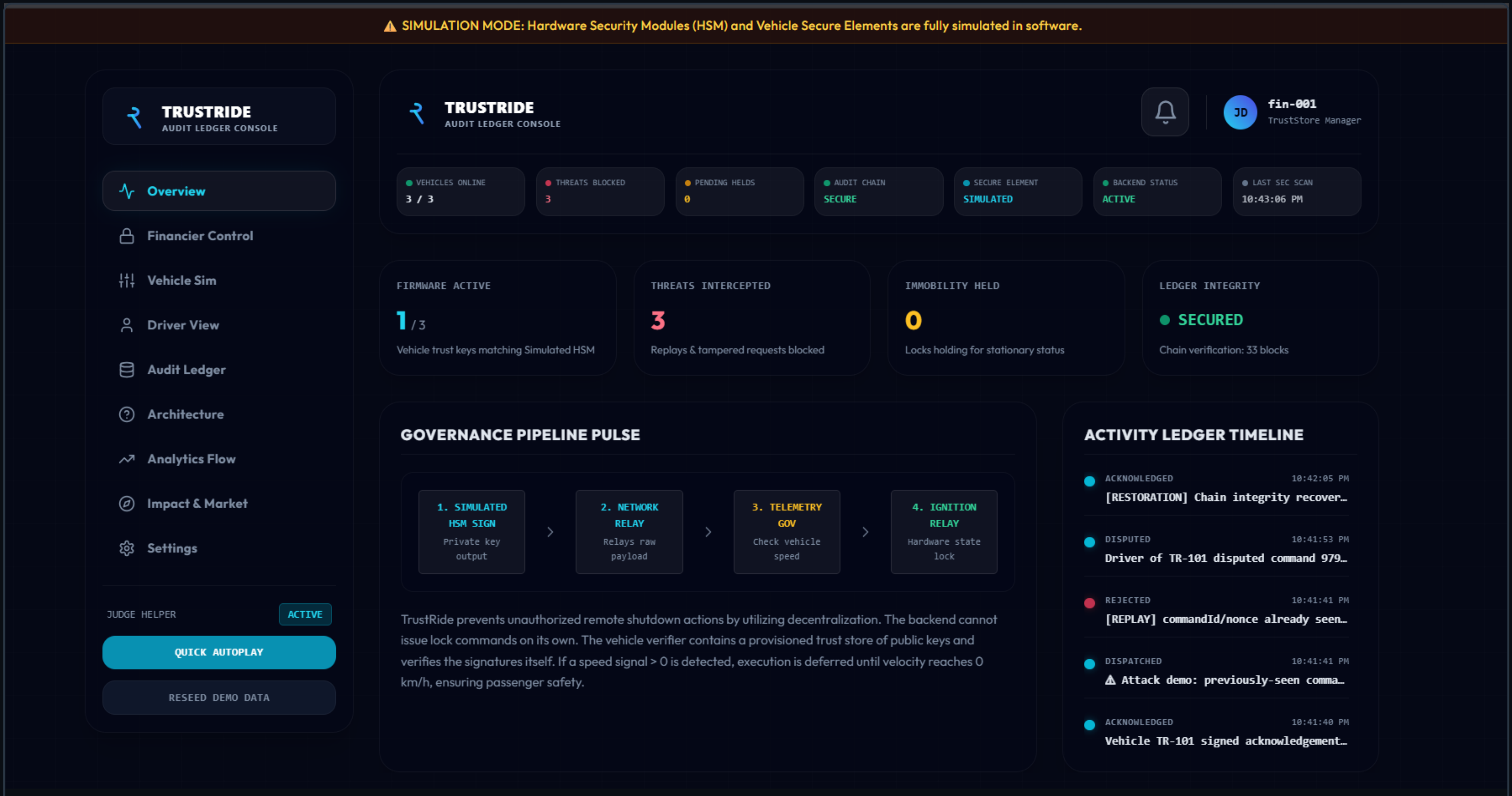
NOT A MOCKUP

A fully working prototype



LIVE CONSOLE

Real signatures. Real hash-chained logs.



REAL-WORLD IMPACT

The scale this protects, in numbers

$1.62B

India's e-rickshaw market in 2026 — growing to $3.14B by 2031

~700K

new e-rickshaws registered in 2024 alone, per VAHAN state data

316K

additional e-rickshaws targeted under the PM E-DRIVE subsidy scheme

Most

financed through NBFCs, not owned outright — each one a live shutdown target today

Every financed e-rickshaw on the road today can already be remotely disabled — with no signature, no motion check, and no record of who or why. TrustRide is the accountability layer that doesn't exist yet.

SCALABILITY

One governance layer, any financed EV

1

E-rickshaws — now

The immediate target: cheapest BMS hardware, least regulation, highest exposure to exactly this exploit class.

2

E-two-wheelers & e-autos

Same BMS/VCU architecture, same NBFC financing model — the verification engine drops in with no redesign.

3

Fleet & last-mile delivery EVs

Four-wheeler EV fleets already use remote-disable telematics — the identical crypto and audit layer applies directly.

The protocol doesn't care what the vehicle is — it cares who's allowed to shut it down, and whether that's provable. That's what scales.